

MAT 524: Functions of a Real Variable II

Homework X

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[[Problem One]]:

Problem Statement

For $f, g \in L^1(\mathbb{R})$, define their convolution by

$$f * g(x) = \int_{\mathbb{R}} f(x-t)g(t)dt.$$

It can be shown that $f * g(x)$ exists for almost all x , and that $f * g \in L^1(\mathbb{R})$. Granting this, prove that

(a) $f * g = g * f$ and **(b)** $\widehat{f * g} = \hat{f}\hat{g}$.

Solution

(a) We have

$$f * g(x) = \int_{\mathbb{R}} f(x-t)g(t)dt. \tag{1}$$

Let $u = x - t$. Then, if we consider x to be fixed, we can write $du = -dt$. Furthermore, we have $t = x - u$.

With this substitution, we have

$$\int_{\mathbb{R}} f(x-t)g(t)dt = \int_{\infty}^{-\infty} f(u)g(x-u)(-du) \tag{2}$$

$$= \int_{-\infty}^{\infty} g(x-u)f(u)du \tag{3}$$

$$= \int_{\mathbb{R}} g(x-u)f(u)du \tag{4}$$

$$= g * f(x). \tag{5}$$

(b) First note that since $f * g \in L^1(\mathbb{R})$, it follows that $\int_{\mathbb{R}} |f * g| < \infty$, i.e., the integral converges absolutely.

The Fourier transform of $f * g$ is

$$\widehat{f * g}(y) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x-t)g(t)dt \right) e^{-2\pi ixy} dx \quad (6)$$

$$= \int_{\mathbb{R}} \int_{\mathbb{R}} f(x-t)g(t)e^{-2\pi ixy} dt dx. \quad (7)$$

Let $u = x - t$. Then $x = u + t$ and $du = dx$. Then we have, by Fubini's Theorem,

$$\int_{\mathbb{R}} \int_{\mathbb{R}} f(x-t)g(t)e^{-2\pi ixy} dt dx = \int_{\mathbb{R}} \int_{\mathbb{R}} f(u)g(t)e^{-2\pi i(u+t)y} dt du \quad (8)$$

$$= \left(\int_{\mathbb{R}} f(u)e^{-2\pi iuy} du \right) \left(\int_{\mathbb{R}} g(t)e^{-2\pi ity} dt \right) \quad (9)$$

$$= \hat{f}(y)\hat{g}(y). \quad (10)$$

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